Adaptive Multi-Agent Malware Investigation

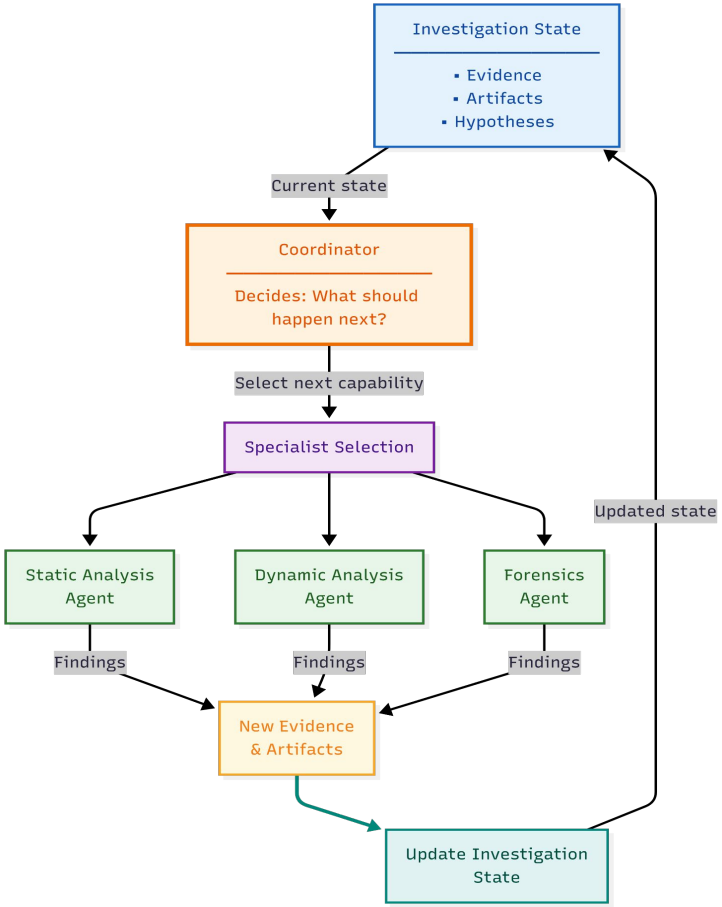
Coordinating Specialized Agents for Dynamic Malware Investigation

From Analysis Pipeline to Adaptive Investigation

Evidence-driven investigation loop

Core Idea: The system does not decide the entire investigation upfront. It continuously decides what should be investigated next based on the evidence discovered so far.

Fixed Pipeline	Adaptive Investigation
Predefined analysis order	Evidence determines next step
Each stage runs once	Analysis can be revisited
Focus on analysis tasks	Focus on investigation hypotheses
Limited feedback between stages	Continuous feedback through shared state
Same workflow for every sample	Investigation path adapts to each sample



Architecture [ref]

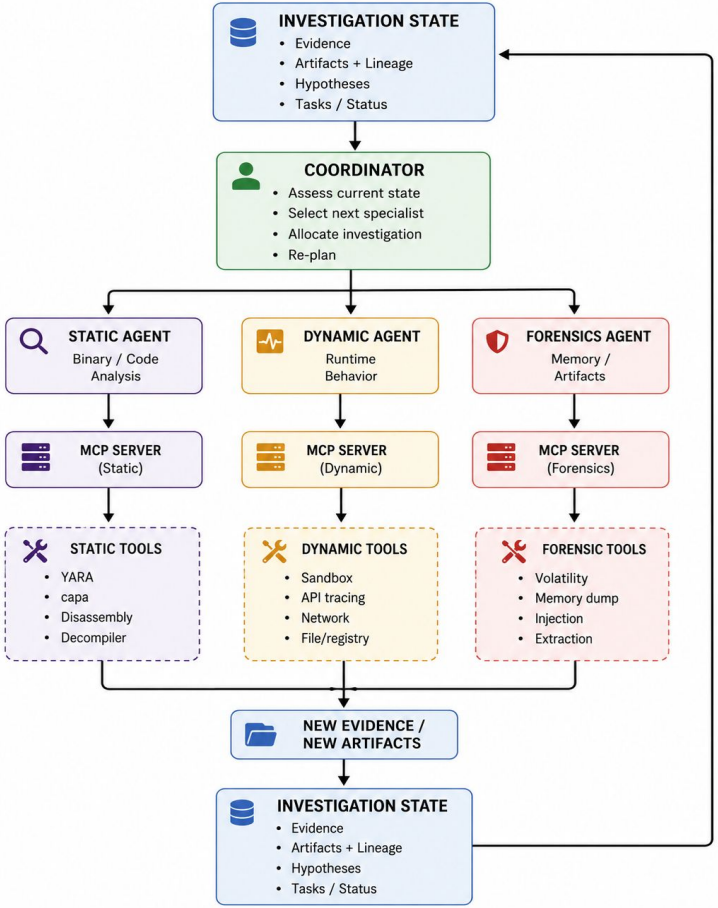
The architecture has 4 layers

1. Investigation Layer
The **Coordinator** manages the global investigation strategy.

2. Specialist Layer
Static, Dynamic, and Forensics agents provide different forms of investigative expertise.

3. Capability Layer
MCP exposes analysis tools to each specialist without coupling the agents directly to individual tools.

4. State Layer
Evidence, artifacts, hypotheses, and provenance form the shared investigation state that drives subsequent decisions.



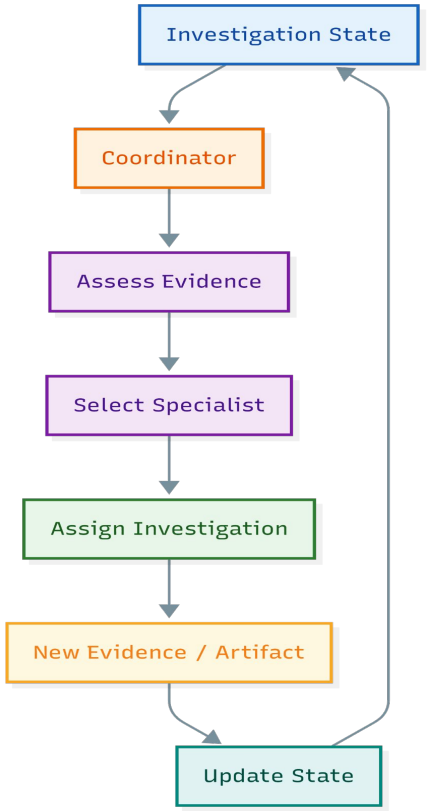
Core Design Principle

Coordinator decides WHO investigates next.
Specialist decides HOW to investigate.
Evidence determines WHAT happens next.

Two-Level Reasoning: Global Coordination + Local Investigation

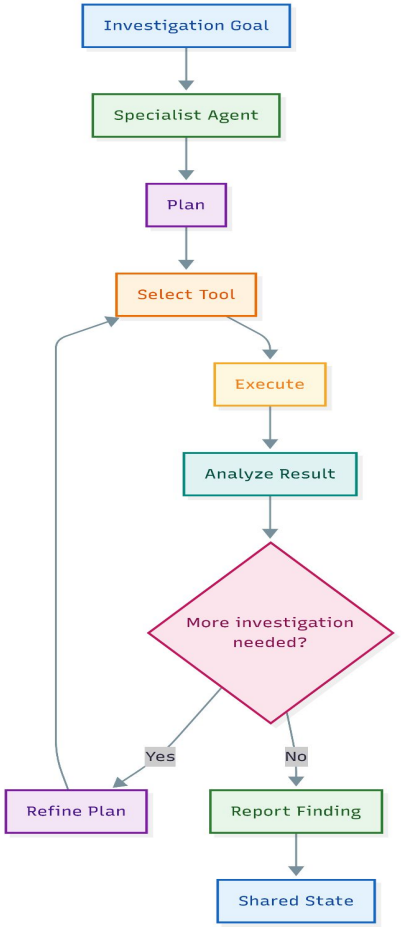
Global Investigation Loop

Coordinator decides: *What should happen next?*



Local Specialist Loop

Specialist decides: *How should I investigate it?*

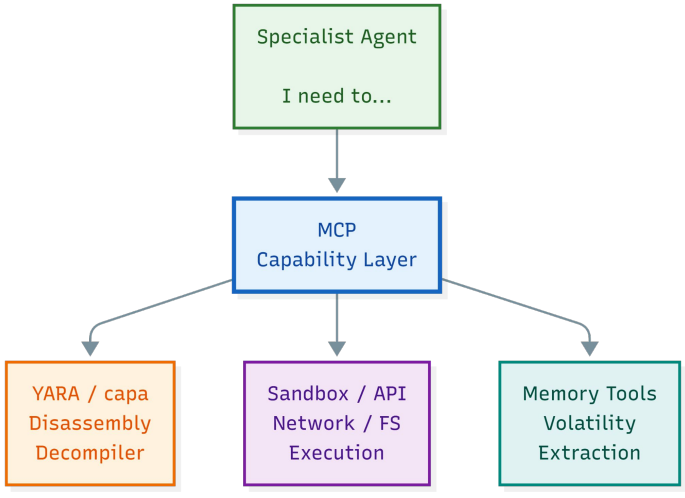


MCP as the Capability Layer

Separating Reasoning from Tool Execution

The MAS should **not be tightly coupled to individual malware-analysis tools**.

Instead, specialists interact with tools through **MCP-based capability servers**.



Separation of Responsibilities

Agent → Reasoning

- Understands the investigation objective
- Forms hypotheses
- Selects appropriate capabilities
- Interprets results

MCP → Capabilities

- Exposes tools through standardized interfaces
- Validates tool inputs
- Executes analysis operations
- Returns structured results

This gives us:

- **Modularity** — add or replace tools without redesigning agents
- **Extensibility** — new analysis capabilities become available to specialists
- **Isolation** — tool execution can be separated from agent reasoning
- **Auditability** — tool calls and results can be recorded
- **Specialization** — each agent accesses the capabilities relevant to its domain

Shared Investigation State

The Memory of the Investigation

What the State Captures

ARTIFACTS & LINEAGE
What was discovered and where it came from.

```
malware.exe → decrypted payload → payload.bin → shellcode
```

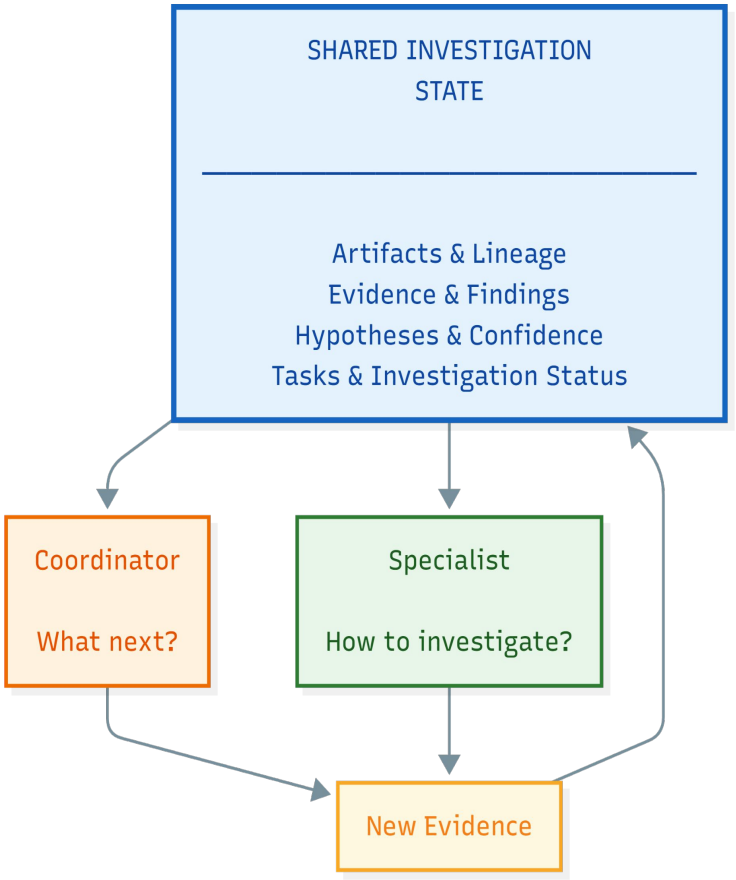
EVIDENCE & FINDINGS
Observations produced by tools and agents.

> “Payload is decrypted in memory.”

HYPOTHESES & CONFIDENCE
What the system currently believes — and how strongly.

> **H_1:** Payload contains process-injection capability
> **Confidence:** 0.72

TASKS & STATUS
What has been investigated, what remains unresolved, and what is currently running.



What Is Novel in Our Approach?

From Agentic Malware Analysis to Adaptive Investigation

Existing LLM/MAS work has demonstrated:

- Autonomous malware analysis
- LLM-driven tool selection
- Specialized analysis agents
- Hypothesis verification and multi-agent consensus
- Evidence-based malware detection

Our proposed contribution

Make the evolving investigation — rather than an individual analysis task or final classification — the primary object of adaptation.

Core Novelty

Evidence → Hypothesis → Agent Selection → Investigation → New Artifact/Evidence → Re-planning

The investigation path emerges from what the malware reveals — rather than being prescribed beforehand.

Existing Direction	Our Direction
Tool / task orchestration	Investigation orchestration
Agent performs analysis	Agents dynamically selected based on evidence
Verify existing findings	Update and re-plan investigation hypotheses
Evidence supports final verdict	Evidence drives the next action
Analysis stages largely predefined	Analysis stages can be revisited
Individual outputs	Persistent evidence + artifact lineage + hypotheses